Xiaoqiu (Shelly) Yang

xiaoqiuy@seas.upenn.edu | 2679691149 | 2101 Market Street

Education

University of Pennsylvania, Philadelphia, PA
MSE. in Bioengineering **Sep 2023 – Present**

- **Cumulative GPA**: 3.97/4.00
  - Academic Dean's List **(**Sep 2023 – May 2024)
- **Relevant Coursework**: Vaccine and Immune Therapy, Stem Cells, Proteomics and Drug Delivery, Biomicrofludics

Boston University, Boston, MA
B.A. in Biochemistry and Molecular Biology **Sep 2019 – May 2023**

- **Cumulative GPA**: 3.75/4.00 (Cum Laude, Top 30%)
  - Academic Dean's List (Jan 2020 – May 2023)
- **Relevant Coursework**: Cell Biology, Molecular Biology, Biochemistry II, Microbiology, Organic Chemistry, Genetics

Publications

1. Budbazar, E., Sulser Ponce De Leon, S., Tsukahara, Y., Liu, H., Huangfu, Y., Wang, Y., Seabra, P.M., **Yang, X.**, Goodman, J.B., Wan, X. and Chitalia, V., **(2023)**. Redox Dysregulation of Vascular Smooth Muscle Sirtuin-1 in Thoracic Aortic Aneurysm in Marfan Syndrome. *Arteriosclerosis, Thrombosis, and Vascular Biology, 43(8), e339-e357.* (Published, IF=8.7)

2. Zhou, Y., Hou, D., Marigo, C. C., Bonelli, J., Rocas, P., Cheng, F., **Yang, X.**, ... & Han, J. **(2022)**. Redox-Responsive Polyurethane-Polyurea Nanoparticles Targeting to Aortic Endothelium and Atherosclerosis. *Iscience, 25(11).* (Published, IF= 6.107)

3. Rusinkevich, Y., Stachelek, S., Hall, E., Fishbein, L., Lu, P., Peng, T., Toomey, N., **Yang, X.**, Pressly, B., …, Levy, R. The Role of PIEZO1 in Serotonin Mediated Progression of Mitral Regurgitation. *Science Advances.* (Under Review, IF=11.7)

Research Experience

Dr. Emanuela Bruscia's Lab **Jun 2024 – Aug 2024**
Yale University *New Haven, CT*

- <u>Independent: Tracing CCR2$^+$ Monocytes Migration and Differentiation in Lung Tissues during Chronic Inflammation</u>
  - Administered tamoxifen to induce fluorescent reporter expression in the CCR2-CreER-GFP:mCherry mouse model.
  - Established chronic lung inflammation via repeated LPS nebulization over time to study monocyte differentiation.
  - Prepared single-cell suspensions from lung, blood, bronchoalveolar lavage fluid (BALF), and bone marrow tissues.
  - Quantified CCR2$^+$ populations using a 12-color panel via flow cytometry.

- <u>Directed: Hematopoietic Stem Cells Exhaustion and Myeloid Skewing Induced by Chronic Inflammation</u>
  - Utilized wild-type (WT) and cystic fibrosis transmembrane conductance regulator knockout (CFTR-KO) mice to investigate the effects of chronic LPS-induced inflammation on immune cell populations.
  - Quantified distinct myeloid and lymphocyte subpopulations in lung, blood, and bone marrow cells using multi-channel flow cytometry; Evaluated cell population modifications under chronic LPS-induced inflammation.

- <u>Directed: Testing Thiol Compound-Mediated Modulation of Inflammation in Macrophages</u>
  - Conducted complete blood count (CBC) to assess leukocytes, thrombocytes, erythrocytes, and their subpopulations.
  - Quantified the proinflammatory cytokine response to thiol compound I-152 with ELISA assays.

Dr. Robert J. Levy's Lab **Sep 2023 – Present**
University of Pennsylvania *Philadelphia, PA*

- <u>Thesis Project: Piezo1 Activaton Stimulates Serotonin Receptor Signaling Mediated Progression of Mitral Regurgitation</u>
  - Maintained human mitral valve interstitial (MVIC) and endothelial (MVEC) cell lines derived from healthy individuals and patients with mitral regurgitation (MR) to study cellular responses under pathological conditions.
  - Utilized the FlexCell Bioreactor System to expose MVICs to static and oscillating flow for 24 hours, modeling the mechanical environment associated with mitral regurgitation.
  - Conducted RNA extraction and reverse transcriptase quantitative PCR (RT-qPCR) to assess the impact of Piezo1-mediated signaling on serotonin (5HT)-related gene expression.
  - Investigated the role of Piezo1 by employing siRNA knockdown techniques in static culture, comparing effects on 5HT-related gene expression between Piezo1 knockdown and non-coding RNA controls.
  - Examined the release of Pro-collagen to extracellular space with ELISA assays.

Dr. Jingyan Han's Lab **Sep 2019 – May 2023**
Boston University *Boston, MA*

- <u>Directed: Redox Dysregulation of Vascular Smooth Muscle Sirtuin-1 in Thoracic Aortic Aneurysm in Marfan Syndrome</u>
  - Evaluated the activity of extracellular matrix-degrading enzymes MMP2 and MMP9 using in-gel zymography to assess the degree of aortic wall remodeling.
  - Examined the protein-S-glutathionylation levels in normal and Thoracic Aortic Aneurysm-prone mice using immunofluorescence.

- <u>Directed: Role of protein-S-glutathionylation in endothelial dysfunction and atherosclerosis</u>
  - Quantified glutationylated protein via western blot to explore the role of S-glutathionylation in endothelial dysfunction.

- Directed: Maintain mouse model for Redox-responsive Puua-NCs targeting atherosclerotic lesions
  - Created shared Excel sheets for mouse colonies; Maintained mouse models through breeding, weaning, and genotyping.

Research Symposium Poster Presentations

Poster Presentation, 18th Society of Chinese Bioscientists in America — Jul 2022
Tufts University — *Boston, MA*
Poster Presentation, Evans Day, Boston University — Oct 2022
Hiebert Lounge, Boston University Medical Campus — *Boston, MA*
Poster Presentation, Undergraduate Research Opportunities Symposium, Boston University — Oct 2022
Charles River Campus, Boston University — *Boston, MA*

- Presented research project on the detrimental effects of alcohol on vascular health in hypercholesterolemic populations, focusing on redox signaling pathways, including glutathionylation and the role of glutaroxin with related preliminary data.

Honors and Awards

Cystic Fibrosis Foundation (CFF) Student Traineeship Award — Jun 2024 – Sep 2024
Yale University — *New Haven, CT*

- Developed a research proposal to track $CCR2^+$ cells in response to chronic lung inflammation.
- Awarded summer research stipend from the Cystic Fibrosis Foundation after the proposal was approved.

Undergraduate Research Opportunities Award ($5700 Awarded) — May 2022 – Aug 2022
Boston University — *Boston, MA*

- Developed a research proposal and assisted with experiments on thoracic aortic aneurysm project during the summer.

Proficiency, ASBMB Certification Exam — Aug 2023
Boston University — *Boston, MA*

- Recognized for proficiency in Biochemistry and Molecular Biology through the national ASBMB Certification Exam.

Young Investigator Poster Award, **2nd Annual Spring Symposium of Society of Chinese Bioscientists in America** — Feb 2023
Karp Boardroom, Boston Children's Hospital — *Boston, MA*

- Submitted an abstract that was selected for presentation in a trainee session; Presented research in an oral presentation, answered questions in the Q&A, and was awarded the Young Investigator Poster Award along with a $100 monetary prize.

Junior Organizer, 2nd Annual Spring Symposium of Society of Chinese Bioscientists in America — Feb 2023
Folman Auditorium, Enders Building, Boston Children's Hospital — *Boston, MA*

- Printed, assembled, and distributed name tags for director, PIs, and trainees; Actively assisted with questions regarding event details and logistics; Ordered and coordinated the set up of breakfast, lunch and refreshments.

Activities and Outreach

Boston Children's Hospital Volunteer — **Aug 2022 – May 2023**
Boston Children's Hospital — *Boston, MA*

- Volunteered 4 hours weekly in the Cardiology Inpatient Unit, organizing books and toys for pediatric patients.
- Provided comfort to children through snuggling and storytelling, while offering companionship and support to parents.

Boston University Lab Mentor — **July 2022 – May 2023**
Boston University — *Boston, MA*

- Trained a student in genotyping, western blot, and data analysis, explaining the underlying mechanisms behind each step.

GABE-BMES Mentorship — **2024 – 2025**
University of Pennsylvania — *Philadelphia, PA*

- Met with undergrads at Penn Biomedical Engineering Society to offer advice on career and graduate school paths.

Boston University International Peer Mentor — **Aug 2021 – May 2022**
Boston University — *Boston, MA*

- Supported 5 international students in academic and social adaptation through, weekly check-ins and resource connections
- Facilitated students' transition to American culture by coordinating mentorship activities.

Clinical Observation — **Dec 2021**
New England Eye Center (NECO) — *Boston, MA*

- Shadowed ophthalmologists and observed patient examinations; Learned about patient communication and the workflow in a clinical setting how to cater to patients professionally and how to effectively interact with patients.

Skills

In Vitro Molecular & Cell Biology: RNA extraction, RT-qPCR, chemiluminescent and fluorescent western blotting, agarose and polyacrylamide gel electrophoresis, flow cytometry, ELISA, In-Gel Zymography, Immunofluorescence, cell culture, siRNA Transfection for Gene Knockdown, cholesterol and triglyceride measurement, BCA and Bradford protein assays
In Vivo Animal models: Mouse Model Development and Maintenance, Intraperitoneal Drug Injection, Chronic Inflammatory Disease Model (LPS Neubulization), Mouse genotyping
Data Analysis & Visualization: FlowJo, Graphpad, Image J, R, Microsoft Office Suite, Adobe Photoshop, BioRender